

Mohamed Saad

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EDUCATION

Kafr-Elsheikh University

Kafr Elsheikh, EGY

Bachelor's Degree in Mechatronics Engineering

July 2023

PROFESSIONAL SUMMARY

Innovative Mechanical Design Engineer with expertise in transformers, oil & gas equipment, and advanced mechanical systems. Skilled in CAD modeling, FEA, and end-to-end design workflows, with proven success in API-standard rig projects, reverse engineering, and precision design of transformer assemblies. Known for cross-team collaboration and delivering reliable, high-quality solutions.

EXPERIENCE

ELSEWEDY ELECTRIC

Mar 2024 – Present

Mechanical design engineer (R&D)

10th of Ramdan, EGYPT

- Design parametric models to accelerate transformer development process.
- Create standardized drawings to support production and quality consistency.
- Perform FEA and mechanical testing for design validation.
- Prepare accurate bill of materials for active part assemblies.
- Collaborate across teams to integrate mechanical and control systems.

Energy Shields

Dec 2024 – Mar 2025

Mechanical design engineer (R&D)

Maadi, EGYPT

- Delivered projects for ADES and Sino-Tharwa with API certified rig builder.
- Led R&D initiatives to develop and enhance rig components and structural designs.
- Reverse engineered rig components for maintenance and refurbishment.
- Designed steel structures including masts and setback areas.
- Produced CAD models and fabrication drawings to API standards.
- Ensured structural integrity through cross team engineering collaboration.

KEY PROJECTS:

- ADES – Designed a custom toolbox for transporting drilling pipes to the rig.
- NAPSCO – Built and reverse-engineered a complete mud system, and carried out setback area maintenance including structural repairs and refurbishment.
- SinoTharwa – Maintenance of setback area, including full dismantling, structural rebuild, and creation of an as-built model.

- Delivered 200+ mechanical design projects using advanced CAD tools.
- Designed and manufactured CNC machines and 3D printers.
- Created wiring diagrams and supervised control panel assembly.
- Managed full design lifecycle from concept to production.
- Collaborated with teams to solve design and system issues.

KEY PROJECTS:

- Designed advanced production lines, sheet-metal lockers, and plastic molds for efficient mass manufacturing.
- Engineered innovative consumer products including washing machines, choppers, microwaves, and a mini-bus with integrated coffee equipment.
- Developed safety and utility concepts such as an anti-collision helmet, mobile gas station, and mobile ICU.
- Created specialized mechanisms and vehicles, from umbrella systems inspired by Makkah's Haram to military-based armed vehicle designs.

TECHNICAL SKILLS

Software: Solidworks, AutoCAD, Inventor, Fusion 360, Solid Edge, Ansys, Automation studio

Languages: Python, MATLAB, PLC, C, C++, C#, English.